

Spatial Equilibrium, Trophic Cascade Refinement, and Thermodynamic Ledger Stabilization in Artificial Ecosystems

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Sprint 028 Technical Preprint

Abstract

Ecosystems are quintessential non-equilibrium thermodynamic systems that maintain internal organization through continuous energy dissipation and matter recycling. We present the formalization and architectural implementation of Sprint 028 for the *Web of Life* platform. This sprint establishes a rigorous coupling between spatial nutrient diffusion, organismal movement energetics, multi-tier trophic interactions (including detritivory), and real-time thermodynamic accounting. By enforcing strict mass-energy conservation (First Law) and monotonic entropy accumulation (Second Law) across discrete monad stock transitions, our framework eliminates numerical drift. Automated verification confirms mass discrepancy bounds of $< 10^{-12}$ units over extended simulation horizons, stabilizing predator-prey phase trajectories and advancing mechanistic fidelity in artificial life simulations.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

1 Introduction and Theoretical Foundations

Previous iterations of the *Web of Life* engine established localized metabolic accounting and solar input bounds (RFC 027). However, spatial decoupling and unconstrained carcass mineralization introduced subtle mass-energy discrepancies and numerical instabilities during trophic cascades.

Sprint 028 addresses these limitations by introducing:

1. **Spatial Nutrient Transport:** Continuous-space diffusion coupled with discrete biome patches.
2. **Strict Matter Conservation:** Zero-loss organismal death and detrital scavenging.
3. **Dynamic Thermodynamic Ledgering:** Real-time tracking of heat dissipation and global entropy accumulation (ΔS).

2 Physical and Biological Process Formalization

2.1 Spatial Nutrient Transport

Nutrient availability for species $i \in \{C, N, P\}$ within a `BiomePatch` at coordinate x is governed by the reaction-diffusion equation:

$$\frac{\partial N_i(x, t)}{\partial t} = D_i \nabla^2 N_i(x, t) + R_i(x, t) - U_i(x, t) \quad (1)$$

where D_i represents the diffusion coefficient, $R_i(x, t)$ is the mineralization and replenishment rate, and $U_i(x, t)$ is the biological uptake rate by producers and consumers.

2.2 Thermodynamic Laws and Entropy Generation

In accordance with classical thermodynamics:

$$\text{First Law (Mass Conservation): } \Delta M_{\text{total}} = \Delta M_{\text{organisms}} + \Delta M_{\text{environment}} = 0 \quad (2)$$

$$\text{Second Law (Entropy Increase): } \Delta S = \sum \frac{Q_{\text{dissipated}}}{T_{\text{ambient}}} \geq 0 \quad (3)$$

where $Q_{\text{dissipated}}$ is metabolic heat loss in joules and T_{ambient} is the local ambient temperature in Kelvin.

3 Systems Architecture and Monad Transitions

The simulation extends existing abstract hierarchies to support specialized ecological roles:

- **SpatialNode / BiomePatch:** Manages localized carrying capacities and elemental stocks (C, N, P, Water).
- **Detritivore:** Implements specialized monad stock transitions for scavenging dead biomass, ensuring zero matter loss upon organismal death with ecological efficiency $\eta = 0.15$.

Table 1: Monad State Transition Table and Thermodynamic Constraints

Source State	Event / Action	Target State	Thermodynamic Constraint
LiveOrganism	MetabolicProcess	DepletedOrganism + Heat	$\Delta M = 0, \Delta S > 0$
LiveOrganism	Death	Carcass	100% stock transfer (C, N, P)
Carcass	Scavenge	Assimilated Biomass + Residue	Mass conservation ($\eta = 0.15$)

4 Verification and Acceptance Criteria

Sprint 028 establishes rigorous verification gates:

1. **Conservation Test:** Total atomic mass varies by $< 10^{-12}$ units over 1,000 simulation ticks.
2. **Entropy Growth Test:** Global entropy $\sum \Delta S$ remains monotonically non-decreasing.
3. **Trophic Balance:** Predator-prey oscillations exhibit stable limit cycles without numerical crashes.

5 Conclusion

Sprint 028 successfully closes the thermodynamic feedback loop within the *Web of Life* simulation engine. By rigorously enforcing spatial equilibrium, strict mass conservation, and entropy ledger stabilization, we provide a robust foundation for emergent macroecological phenomena.

Source Code & Implementation: Available at <https://github.com/pascalranoroarijaona/WebOfLife>.